

VOLUME IV - REFERENCES / RESEARCH PAPERS

Research Papers Index

Annotated Bibliography of Academic and Technical Literature

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Programme	Safe & Continuous Haulage under Low-Visibility Monsoon Conditions
Site of Application	Bailadila Iron Ore Mines - Kirandul & Bacheli Complexes, Dantewada, Chhattisgarh
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ABSTRACT

An annotated index of the academic and technical literature underpinning this research package, organised by theme. Each entry records the source, what it establishes, and where in the package it is relied upon. Entries are described in summary form; readers requiring the substance should consult the originals, which are identified sufficiently to be located. Where a claim in the package rests on simulation rather than field measurement, this index says so.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. How to Use This Index

Entries are grouped by theme and carry a reference code of the form RP-nn. Each entry states:

- ? **Source** - author or publishing body, title in summary form, and venue or platform where known.
- ? **Establishes** - what the work contributes to this package, in the package's own words rather than the source's.
- ? **Used in** - the document and section of this package that relies on it.
- ? **Evidence class** - one of: **Field** (measured in real conditions), **Chamber** (controlled artificial conditions), **Simulation** (modelled), **Survey** (review of other work), **Industry** (vendor or operator disclosure).

The evidence class column is the most important feature of this index. A substantial part of the published work on sensing in fog is simulation or fog-chamber based. That is adequate to establish which modality is correct and inadequate to predict detection range at Bailadila. This distinction is carried through the package and is the reason requirement AR-02 exists.

2. Adverse-Weather Perception and Sensor Physics

RP-01. Survey of perception and sensing for autonomous vehicles under adverse weather conditions. Published as a review article in the remote sensing and photogrammetry literature. *Establishes:* comparative attenuation of radar and LiDAR under rain and fog, including the finding that 77 GHz radar attenuation in 25 mm/hr heavy rain is approximately 10 dB/km against approximately 35 dB/km for 905 nm LiDAR under comparable low-visibility conditions; and reported simulation results indicating detection ranges of approximately 260 m in heavy fog, 460 m in mild fog and over 600 m in heavy rain for a linear FM 77 GHz and 1550 nm configuration at a 20 dB SNR threshold. *Used in:* Vol. I Doc. 1 Sec. 3.3; Vol. II Doc. 2 Sec. 2.1; Vol. III Doc. 1 Sec. 4.2. *Evidence class:* Survey, incorporating Simulation.

RP-02. Comparative analysis of LiDAR, radar and camera sensors for autonomous perception: principles, challenges and fusion strategies. *Establishes:* the general superiority of radar in occlusion, dust, glare and precipitation, and the corresponding weakness of LiDAR in fog, rain and smoke together with its higher cost relative to radar or camera. *Used in:* Vol. II Doc. 2 Sec. 2.2. *Evidence class:* Survey.

RP-03. Millimetre-wave radar and vision fusion for object detection in autonomous driving: a review. *Establishes:* mmWave radar detection of obstacles within approximately 250 m against LiDAR within approximately 150 m; and relative velocity measurement by Doppler at approximately 0.1 m/s resolution. *Used in:* Vol. III Doc. 1 Sec. 1, Sec. 2.2. *Evidence class:* Survey.

RP-04. Experimental evaluation of 79 GHz and 300 GHz radar performance in fire environments. *Establishes:* measured propagation of 79 GHz and 300 GHz signals through flame, smoke and steam, with attenuation compared against theory for the steam case; confirmation that radar renders objects visible in these obscurant conditions; and the observation that higher operating frequency reduces antenna aperture for a given beamwidth, with a corresponding size and mass benefit. Also notes that frequencies above 100 GHz remain insufficiently studied. *Used in:* Vol. II Doc. 2 Sec. 2.1; Vol. III Doc. 1 Sec. 3, Sec. 4.2. *Evidence class:* Field/Chamber (controlled fire environments).

RP-05. Comparative treatments of LiDAR versus radar sensing principles for industrial and automotive applications. *Establishes:* wavelength contrast - LiDAR at 0.9 to 1.55 μm against radar at 3 mm to 30 cm - and the consequent difference in behaviour through fog, cloud, precipitation and dust; radar's cost advantage of roughly three to five times for an equivalent sensor; LiDAR's advantage in 3D mapping and high-resolution obstacle detection. *Used in:* Vol. I Doc. 1 Sec. 3.3; Vol. II Doc. 2 Sec. 2.2. *Evidence class:* Industry/Survey.

3. Radar Technology

RP-06. Design of a high-performance 77 GHz millimetre-wave radar, signal processing and control system for autonomous vehicle navigation. *Establishes:* radar front-end and intermediate-frequency design; discrimination of

pre-placed target beacons from other reflectors using reflected-signal polarisation; determination of reflected signal power at incremental distances; and the use of range and bearing measurements to update a Kalman filter estimating vehicle location. *Used in:* Vol. III Doc. 1 Sec. 2, Sec. 8. *Evidence class:* Field (prototype).

RP-07. Radar detection rate comparison via a mobile robot platform at a proving ground. *Establishes:* radar's better performance in low-light and harsh environments alongside its inferior object classification and detection resolution relative to other modalities, and the consequent necessity of multi-sensor data fusion in the automated driving domain; also reports centimetre-level distance precision achievable at 24 GHz over short ranges. *Used in:* Vol. III Doc. 1 Sec. 2.3; Vol. III Doc. 2 Sec. 8. *Evidence class:* Field.

RP-08. Literature on micro-Doppler signature analysis for pedestrian and target classification in millimetre-wave radar. *Establishes:* that limb motion in human gait produces a characteristic time-varying Doppler modulation distinguishable from the single Doppler component of a rigid body, supporting radar-only pedestrian classification. *Used in:* Vol. III Doc. 1 Sec. 10; Vol. III Doc. 3 Sec. 2.2. *Evidence class:* Field.

RP-09. Literature on MIMO radar array synthesis, angular resolution and cascaded imaging radar architectures. *Establishes:* the virtual array relationship between transmit and receive element counts and the resulting angular resolution, on which the lane-discrimination analysis of this package depends. *Used in:* Vol. III Doc. 1 Sec. 2.3, Table 2.2. *Evidence class:* Survey.

RP-10. Literature on automotive radar mutual interference characterisation and mitigation, including chirp randomisation and in-chain interference detection. *Establishes:* the mechanisms of radar-to-radar interference at high device density and the standard mitigations. *Used in:* Vol. III Doc. 1 Sec. 6.3. *Evidence class:* Survey/Simulation.

4. Sensor Fusion

RP-11. Radar and LiDAR sensor fusion in low-visibility environments; and related Monte Carlo framework work coupling radiative-transfer simulation of FMCW LiDAR with a 77 GHz radar link-budget formulation within a common fog microphysical environment. *Establishes:* through integrated beat-spectrum and range-Doppler synthesis, the visibility-dependent degradation, clutter evolution and cross-sensor complementarity; an adaptive fusion kernel balancing LiDAR spatial precision against radar penetration as visibility falls; and the headline result that across simulated visibilities of 0.1 to 2.0 km, **LiDAR alone fails beyond approximately 70 m in dense fog while the fused system sustains detection beyond 100 m** at a probability of detection of 0.9 and false alarm probability of $1e-3$. *Used in:* Vol. II Doc. 1 Sec. 7; Vol. II Doc. 2 Sec. 2.2; Vol. III Doc. 1 Sec. 4.2; Vol. III Doc. 2 Sec. 4, Sec. 8; Vol. III Doc. 3 Sec. 3. *Evidence class:* **Simulation**. This is the most heavily relied-upon quantitative result in the package and it is modelled, not measured. It establishes the direction and approximate magnitude of the radar-LiDAR divergence in fog; it does not establish Bailadila performance.

RP-12. RobustSENSE project architecture integrating LiDAR with long-range 77 GHz and short-range 24 GHz radar, stereo and thermal cameras, connecting a LiDAR detection layer to a performance assessment layer. *Establishes:* the architectural pattern in which supplementary sensor data is used by the vehicle control layer when LiDAR performance is assessed as degrading below a critical level; tested with roadside LiDAR in foggy airport conditions at low hardware cost (under EUR 1000 for the components concerned). *Used in:* Vol. II Doc. 1 Sec. 7; Vol. III Doc. 2 Sec. 8; Vol. III Doc. 3 Sec. 3. *Evidence class:* Field (roadside trial).

RP-13. Multimodal Vehicle Detection Network (MVDNet) fusing LiDAR and radar, extracting features and generating proposals from both sensors before region-wise multimodal fusion. *Establishes:* substantially better detection performance than LiDAR alone in fog, evaluated on a dataset derived from the Oxford Radar RobotCar data. *Used in:* Vol. II Doc. 1 Sec. 7; Vol. III Doc. 3 Sec. 3. *Evidence class:* Simulation/derived dataset.

RP-14. Mechanical pivoting radar fused with LiDAR for SLAM in a low-visibility fog environment. *Establishes:* that a radar with only a 15 m measurement range nonetheless possesses sufficient fog penetration to sustain landmark searching where LiDAR fails, demonstrated on a robot platform. *Used in:* Vol. II Doc. 1 Sec. 7; Vol. III Doc. 1 Sec. 8. *Evidence class:* Field (robot platform).

5. Positioning and Navigation

RP-15. Literature on tightly coupled GNSS/INS integration, error-state Kalman filtering and non-holonomic constraint aiding for land vehicles. *Establishes:* the benefit of tight over loose coupling under partial satellite visibility, and the

drift suppression achieved by non-holonomic constraints and odometric aiding. *Used in:* Vol. III Doc. 2 Sec. 1.5, Sec. 1.7. *Evidence class:* Survey/Field.

RP-16. Literature on RTK carrier-phase ambiguity resolution, baseline-length dependence and multipath as the dominant non-differenced residual error. *Establishes:* accuracy expectations by correction technique and the fact that multipath is not removed by differential correction, which drives the pit-environment assessment in this package. *Used in:* Vol. III Doc. 2 Sec. 1.2, Sec. 1.3. *Evidence class:* Survey/Field.

RP-17. Literature on receiver autonomous integrity monitoring, advanced RAIM and protection-level computation. *Establishes:* the protection-level concept - a bound on position error to a stated probability - transferred in this package from aviation practice to the lane-guidance alert limit. *Used in:* Vol. III Doc. 2 Sec. 1.8. *Evidence class:* Survey.

RP-18. Literature on equatorial ionospheric scintillation and its effect on GNSS carrier tracking in the Indian region. *Establishes:* elevated scintillation risk at low magnetic latitudes and the consequent value of short RTK baselines and robust reacquisition. *Used in:* Vol. III Doc. 2 Sec. 1.4. *Evidence class:* Field.

6. V2X and Cooperative Systems

RP-19. Literature on C-V2X PC5 sidelink Mode 4 and NR-V2X Mode 2 autonomous resource selection, and comparative studies of C-V2X against ITS-G5 under load. *Establishes:* the semi-persistent scheduling mechanism and its more graceful degradation under density relative to CSMA contention, which underpins the bearer recommendation of this package. *Used in:* Vol. III Doc. 4 Sec. 3.2, Sec. 3.5. *Evidence class:* Simulation/Field.

RP-20. Literature on decentralised congestion control in vehicular networks, including adaptive broadcast rate and transmit power control. *Establishes:* the mitigation set applied in the scalability analysis of this package. *Used in:* Vol. III Doc. 4 Sec. 5. *Evidence class:* Simulation.

RP-21. Literature on sub-GHz versus 5.9 GHz propagation, diffraction loss dependence on frequency, and non-line-of-sight performance in obstructed terrain. *Establishes:* the physical basis for the package's caution that a bearer with superior specifications may be inferior in ridge-and-switchback terrain. *Used in:* Vol. III Doc. 4 Sec. 4. *Evidence class:* Survey/Field.

RP-22. Patent and technical literature on mine collision detection and avoidance systems, including tiered proximity thresholds - safe, hazardous and critically hazardous distance thresholds - and integration of personnel collision detection units with existing safety equipment such as cap lamps, together with mine operations analytics and monitoring architectures receiving data from mobile objects and other mine devices. *Establishes:* the graded-threshold pattern and the cap-lamp-integrated personal device form factor adopted in this package. *Used in:* Vol. I Doc. 3 FR-21, FR-23; Vol. III Doc. 4 Sec. 7. *Evidence class:* Industry.

7. Mining Safety and Collision Avoidance

RP-23. Survey of collision avoidance systems for underground mines: sensing protocols. *Establishes:* the decomposition of collision avoidance into a proximity detection subsystem (visual, ultrasonic, acoustic or electromagnetic sensing), a decision-making subsystem consolidating that information to determine whether a hazard is imminent, and an intervention control subsystem; and the observation that advance inspection of vehicles and infrastructure in the vicinity improves the collision-avoidance success rate while alleviating false alarm problems. *Used in:* Vol. I Doc. 2 Sec. 2.5; Vol. II Doc. 1 Sec. 2.4. *Evidence class:* Survey.

RP-24. Industry reporting on Hexagon MineProtect Collision Avoidance System, versions 10 and subsequent. *Establishes:* consolidation of vehicle hardware into a smart antenna integrating GPS, RF, Wi-Fi, LTE and ultra-wideband time-of-flight, plus a five-inch LCD display; integration of collision avoidance with operator alertness, object detection, personal protection and vehicle intervention on shared hardware; a web-based reporting and analytics platform; subsequent addition of 4D radar support and interoperability with personnel tagging; and deployment scale reported at more than 50,000 vehicles at the CAS 10 announcement and subsequently more than 65,000 units across over 100 open-pit and underground sites. *Used in:* Vol. II Doc. 1 Sec. 2.1; Vol. III Doc. 1 Sec. 5; Vol. III Doc. 4 Sec. 1. *Evidence class:* Industry.

- RP-25.** Industry reporting on SAFEmine collision avoidance characteristics. *Establishes:* a dedicated radio system providing real-time traffic awareness and collision avoidance in open pits, described as offering 360 degree protection at any speed and any visibility with a typical range of 800 m and without requiring line of sight, with more than 18,000 mining vehicles protected at the time of reporting and availability across major OEM haul trucks. *Used in:* Vol. II Doc. 1 Sec. 2.1; Vol. II Doc. 2 Sec. 2.2; Vol. III Doc. 4 Sec. 1. *Evidence class:* Industry.
- RP-26.** Industry reporting on large-scale CAS deployments, including Rio Tinto's Oyu Tolgoi copper-gold open pit in Mongolia (over 200 systems across 290 t class haul trucks, rope and hydraulic shovels, dozers, graders, wheel loaders, drill rigs, service and light vehicles) and Gold Fields' Tarkwa mine in Ghana (over 220 mine vehicles, installed alongside operator alertness, personal alert and tracking radar). *Establishes:* that fleet-wide, mixed-equipment CAS deployment at the scale contemplated for Bailadila is routine industry practice. *Used in:* Vol. II Doc. 1 Sec. 2.1. *Evidence class:* Industry.
- RP-27.** Industry material on ADAS in mining and DGMS enforcement thereof. *Establishes:* the role of collision avoidance systems using radar and LiDAR with ultrasonic and camera support to detect people, vehicles and objects; continuous blind-spot and surrounding-area monitoring with visual and audio alerts; and the observation that large machines frequently operate in close proximity in low-visibility zones. *Used in:* Vol. I Doc. 2 Sec. 2.5; Vol. II Doc. 1 Sec. 2.5. *Evidence class:* Industry.
- RP-28.** Industry material on DGMS-compliant HEMM safety equipment supply in India, covering automatic fire suppression, proximity warning systems, acoustic vehicle alerting systems, reverse camera systems, AI pedestrian detection, audio-visual alarms, amber beacons, speed limiters and emergency shutdown, with proximity warning implemented using radar sensors and UWB ranging. *Establishes:* the composition and capability of the domestic Indian supply base. *Used in:* Vol. II Doc. 1 Sec. 2.5. *Evidence class:* Industry.

8. Machine Learning for Perception

- RP-29.** Literature on point-based neural architectures for unordered point sets (PointNet and successors) and on pillar and voxel encoders for bird's-eye-view detection (PointPillars and successors). *Establishes:* the architectural options for sparse point-cloud detection adapted in this package to radar with Doppler as an input channel. *Used in:* Vol. III Doc. 3 Sec. 2.1. *Evidence class:* Field (benchmark).
- RP-30.** Literature on range-Doppler map neural processing operating on the radar data cube prior to thresholding. *Establishes:* the retention of information discarded by CFAR detection, at higher compute and data-rate cost. *Used in:* Vol. III Doc. 3 Sec. 2.1. *Evidence class:* Field (benchmark).
- RP-31.** Literature on single-image dehazing, atmospheric scattering model inversion and generative image restoration. *Establishes:* the physical model these methods invert, and thereby the basis for this package's conclusion that they cannot recover a signal extinguished below the noise floor at extinction coefficients of 600-1000 per km. *Used in:* Vol. III Doc. 3 Sec. 9. *Evidence class:* Field (benchmark) - cited here to support a negative conclusion about applicability.
- RP-32.** Literature on quantisation-aware training, pruning and operator fusion for embedded neural inference. *Establishes:* the practical means of meeting the edge latency budget derived in this package. *Used in:* Vol. III Doc. 3 Sec. 11. *Evidence class:* Field.

9. Fog Meteorology and Nowcasting

- RP-33.** Literature on the Koschmieder relation and the definition of meteorological optical range at the 5 per cent contrast threshold. *Establishes:* the relation $MOR = 3.0 / \sigma_{ext}$ used throughout this package to convert reported visibility into extinction coefficient, and thereby to characterise the Bailadila operating point at 600-1000 per km. *Used in:* Vol. I Doc. 1 Sec. 3.3. *Evidence class:* Survey (established physics).
- RP-34.** Literature on Mie scattering theory, the size parameter and the transition between Rayleigh and Mie regimes. *Establishes:* the wavelength-to-droplet-size analysis that is the physical foundation of the entire architecture. *Used in:* Vol. I Doc. 1 Sec. 3.3; Vol. II Doc. 2 Sec. 2.1; Vol. III Doc. 1 Sec. 4.1. *Evidence class:* Survey (established physics).
- RP-35.** Literature on orographic cloud formation, advection fog and radiation fog mechanisms, and on the lifting condensation level in monsoon flow over elevated terrain. *Establishes:* the three-mechanism characterisation of Bailadila fog, which drives the requirement that a solution handle persistent, transient and diurnal regimes. *Used in:*

Vol. I Doc. 1 Sec. 3.2. *Evidence class*: Survey.

RP-36. Literature on short-horizon fog nowcasting using local observation networks, gradient-boosted methods, recurrent and temporal convolutional sequence models, and spatio-temporal graph networks; and on probabilistic forecast verification using Brier score and critical success index. *Establishes*: the model options and verification framework adopted for requirement FR-04 and KPI K12. *Used in*: Vol. III Doc. 2 Sec. 5; Vol. III Doc. 3 Sec. 5. *Evidence class*: Field.

10. Human Factors

RP-37. Literature on trust calibration in automation, covering under-trust leading to disuse and over-trust leading to misuse, and on the role of system transparency in calibrating trust. *Establishes*: the twin failure modes that this package treats as risk R5 and failure mode F5, and the design response of persistently displaying effective sensing range and system health. *Used in*: Vol. I Doc. 2 Sec. 6; Vol. III Doc. 3 Sec. 14. *Evidence class*: Field.

RP-38. Literature on alarm fatigue, habituation to frequent alerts, and operator disabling of nuisance-prone safety systems. *Establishes*: the mechanism behind risk R3, rated the highest combined likelihood and impact in the package's risk register. *Used in*: Vol. I Doc. 2 Sec. 6, Sec. 8; Vol. III Doc. 3 Sec. 6. *Evidence class*: Field.

RP-39. Literature on glance-time budgets for in-vehicle displays and on pre-attentive visual processing. *Establishes*: the sub-2-second single-glance constraint adopted as requirement HR-01. *Used in*: Vol. I Doc. 3 Sec. 5. *Evidence class*: Field.

RP-40. Aviation human-factors literature on synthetic and enhanced vision systems, combined vision systems, and the presentation of database-derived terrain to a pilot in degraded visual conditions. *Establishes*: the interface precedent for the synthetic perspective guidance view proposed in this package. *Used in*: Vol. II Doc. 1 Sec. 6.1; Vol. II Doc. 3 Sec. 4. *Evidence class*: Field.

11. Public Datasets

RP-41. Seeing Through Fog. Multimodal automotive dataset combining controlled fog-chamber and real-world adverse-weather data across camera, LiDAR, radar and gated imaging. *Value here*: the closest available proxy for the target conditions; suitable for algorithm development and architecture selection. *Limitation*: automotive scenes, not haul roads; fog chamber droplet distributions differ from monsoon orographic cloud.

RP-42. RADIATE. Radar-focused dataset in adverse weather including fog, rain and snow, with annotated objects. *Value here*: radar detection and tracking development under degraded visibility.

RP-43. Oxford Radar RobotCar. Scanning radar dataset with LiDAR and camera, widely used as the basis for derived adverse-weather benchmarks including the MVDNet evaluation. *Value here*: radar odometry and localisation development, supporting Vol. III Doc. 1 Sec. 8.

RP-44. nuScenes and Waymo Open Dataset. Large-scale multimodal autonomous driving datasets with limited adverse-weather content. *Value here*: baseline detection and fusion development; limited direct relevance to the fog case.

Common limitation, stated once and applying to all four. None of these datasets contains a loaded rear-dump truck on a wet ore-fines ramp inside monsoon orographic cloud at MOR 3-5 m. They are adequate for development and architecture selection and inadequate for validation. This is the basis of the package's recommendation that site data collection at Bailadila be treated as a programme deliverable rather than a convenience, and of requirement AR-02.

12. Evidence Strength Assessment

A summary of how strongly the package's principal claims are supported.

Claim	Evidence	Strength
Visible and NIR optical sensing fails at MOR 3-5 m	Established scattering physics (RP-33, RP-34) plus measured attenuation comparisons (RP-01, RP-05)	Very strong. Physics, not opinion.
mmWave radar is substantially unaffected by fog	Established physics (RP-34); measured rain attenuation (RP-01); obscurant measurements (RP-04)	Very strong in direction; moderate in precise magnitude for monsoon fog.
Radar sustains useful detection beyond 100 m where LiDAR fails beyond ~70 m in dense fog	RP-11	Moderate. Simulation-derived. Directionally reliable; not a Bailadila prediction.
Thermal imaging penetrates fog only partially	Physics (RP-34) plus survey material (RP-01)	Strong.
Cooperative RF provides visibility-independent, non-line-of-sight awareness at several hundred metres	Industry disclosure at large deployed scale (RP-25, RP-24, RP-26)	Strong. Industry evidence, independently corroborated by deployment scale.
Sub-metre positioning with continuity through GNSS outage is achievable on haul trucks	Established GNSS/INS practice (RP-15 to RP-17) plus autonomous haulage production use	Very strong.
Alarm fatigue is the dominant operational failure mode of proximity systems	Human-factors literature (RP-38) plus consistent industry experience	Strong.
Detection performance of mining CAS at MOR below 10 m	None located	Absent. This is the market-wide evidence gap identified as G5.
Radar detection range at Bailadila specifically	None	Absent. Requires the Phase 0 instrumented campaign.

The final two rows are the honest conclusion of this index. The package's architectural reasoning rests on strong physical and industrial evidence. Its quantitative performance predictions for Bailadila rest on nothing, because nobody has measured them. That is not a weakness in the reasoning; it is the reason the package recommends a monsoon-gated instrumented field campaign as the single highest-value next action.